Université de Montréal
Data Science

IFT-6758 – Midterm Exam

2024/10/30

Full Name (Firstname LASTNAME)

Student ID (UdeM matricule)

Instructions

- No external resources are allowed.
- Give brief explanations for your answers. We will award partial credit for good reasoning.
- For questions with a box to fill. Put your answer in the box.
- There is extra paper at the end of the booklet. Make sure to write question and part number for any answers in the additional pages you use.

Notations and reminders:

- $\|x_i - x_j\|_2^2 = \sum_{k=1}^{d}([x_i]_k - [x_j]_k)^2$ = Euclidean distance between points x_i and x_j
- $A \setminus B$ = Set difference = Elements in A but not in B
- $H(X) = -\sum_{i=1}^{d} p(X = x_i) \log(X = x_i)$, entropy of a discrete random variable X.
- $I(X, Y) = \sum_{i,j}^{n} p(X = x_i, Y = y_i) \log\left(\frac{p(X=x_i, Y=y_i)}{p(X=x_i)p(Y=y_i)}\right)$ mutual information (or information gain) between two random variables.
- True positive rate: $= \frac{\#(predicted=+|actualstate=+)}{\#(actualstate=+)}$, Accuracy: fraction of correct prediction in total.

Distribution of Marks

Question	Points	Score
1	2	
2	15	
3	10	
4	14	
5	10	
6	13	
7	11	
8	13	
Total:	88	

1. **Warm-up**
   (a) (1 point) Write your full name and UdeM matricule correctly and legibly (see Fig 1).
   (b) (1 point) (At the end) Put your exam on the right pile (the pile corresponding to the **first letter** of your **first name**)

Full Name (Firstname LASTNAME)

Harry James POTTER

Student ID (UdeM matricule)

	3	1	7	1	9	8	0

Instructions

Figure 1: or how to get one point easily.

2. **Classification**

The email inbox uses a classifier to detect spam. The company estimated that 20% of sent emails were spam. The true positive rate of the classifier is 80%, meaning that 80% of spam emails are detected by the test. Additionally, we know that the accuracy of the classifier is 90%.

To complete the following table, we will assume that 100 emails were sent.

	Predicted Positive	Predicted Negative
True Positive		
True Negative		

Table 1: Number of emails in each situation (out of 100 emails)

(a) (4 points) Complete the four cells of Table 1.

(b) (2 points) What is the true negative rate?

ANSWER (in fraction form): $\frac{\cdot}{\cdot}$ (explain why)

(c) (2 points) An email was flagged as spam (predicted positive) by the classifier. It is from a sender who has **never been flagged** before. What is the probability that the email is genuine spam?

ANSWER (in fraction form): $\frac{\cdot}{\cdot}$ (explain why)

(d) (2 points) An email was **not flagged** by the classifier (predicted negative). It is from a sender who has **never been flagged** before. What is the probability that the email is genuine spam?

ANSWER (in fraction form): $\frac{\cdot}{\cdot}$ (**explain** why)

(e) (2 points) In this context, is it preferable to maximize the true positive rate or the true negative rate? (**explain** why)

(f) (3 points) Name 3 relevant features to consider for spam detection.

3. Anomaly Detection

A researcher analyzes the relationship between the height and the age of trees in a specific forest region and has collected the following data:

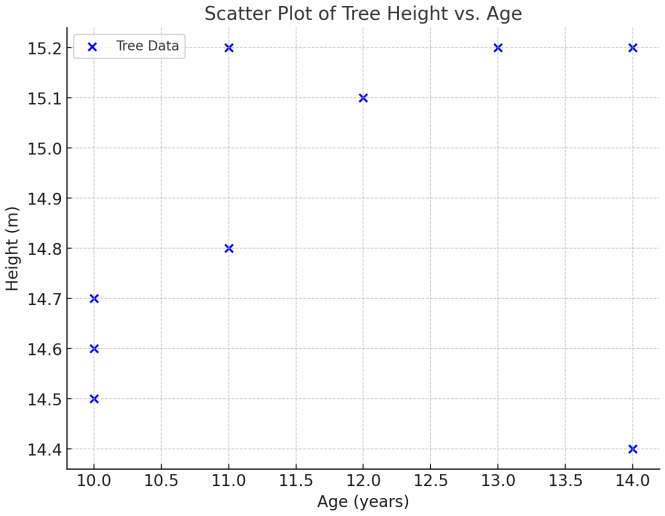


Figure 2: Data Visualization

Height (m)	Age (years)
14.5	10
14.8	11
15.1	12
14.7	10
14.9	13
15.2	11
14.6	10
15.2	14
14.4	14

Table 2: Height and age of trees.

(a) (2 points) Which one of the following facts are **always** true about outliers:

Outliers can significantly affect the empirical mean ☐

Outliers can significantly affect the empirical median ☐

Outliers are easy to identify by visually inspecting the data. ☐

Outliers indicate errors in data collection or entry. ☐

(b) (2 points) Is there an outlier in Figure 2?

ANSWER: Yes ☐, No ☐ (If yes, which one? **And explain** why)

(c) (2 points) Describe the interquartile range (IQR) method for detecting outliers. (Reminder: this method uses the 25th and 75th percentiles).

(d) (2 points) Would the IQR method detect an outlier in the data from Table 2?
ANSWER: Yes ☐, No ☐ (If yes, which one? **And explain** why)

(e) (2 points) Propose another method and explain how it could detect an outlier in this specific case.

4. **Data Visualization and Feature Engineering**

Consider the following dataset, which represents customer data for an e-commerce company. A sample of rows and columns is shown below:

ID	Age	Inc.	Gen.	Pur.Freq	Memb.Yrs	Reg.	Sat.	Pop.	Cust.Score
101	45	75K	M	High	5	NE	Very Satisfied	8M	85
102	30	52K	F	Med	2	W	Satisfied	120K	92
103	62	120K	M	Low	10	S	Unsatisfied	5M	85
104	25	45K	F	High	1	E	Neutral	150K	91
105	55	90K	M	Med	8	NE	Very Satisfied	8M	84

- **ID**: Unique identifier for each customer (identifier)
- **Age**: Age of the customer

- **Inc**: Annual income of the customer in dollars
- **Gen**: Gender of the customer
- **Pur.Freq**: Frequency of customer purchases categorized as "Low", "Medium", or "High"
- **Memb.Yrs**: Number of years the customer has been a member, with a right-skewed distribution (long tail to the right)
- **Reg**: Geographical region of the customer (N, E, W, S, NE, NW, SE, SW)
- **Sat**: Satisfaction level with products purchased, categorized from "Very Satisfied" to "Unsatisfied"
- **Pop.**: Population of the city the customer is from, with significantly more customers from larger cities
- **Cust.Score**: Score of the customer that represents the score they obtained by scratching a scratch card or assigned to them to reveal their lucky prize

(a) (2 points) For the **Reg** column, recommend a technique for representing this feature in the model. Explain your choice based on the values in this column.
Technique [ ] Justify

(b) (2 points) Consider the **Pur.Freq** column. Describe an appropriate encoding technique for this column and explain why this method is suitable.
Technique [ ] Justify

(c) (3 points) The **Inc** column has a wide range of values. Suggest a transformation technique for this column to reduce the effect of large disparities in income.
Technique [ ] Justify

(d) (2 points) It is given that more customers come from larger cities (with higher populations) than smaller ones. What type of plot will you use to visualize the **Pop** feature? Draw an example plot that you would expect for this and label the axes. **Justify** your choice.

(e) (1 point) The **Memb.Yrs** column is right-skewed (there are way more recent customers than older ones). Which transformation or scaling method might be appropriate for this

feature to improve its usability in the model? Justify your answer.
Transform [ ] Justify

(f) (2 points) Some columns may need to be dropped because they are irrelevant or redundant. Describe a visualization technique to identify which features are relevant, irrelevant, or redundant. Explain why this visualization would help determine which features to drop.
Technique [ ] Justify

(g) (2 points) Draw an example plot to depict how much the feature *Cust.Score* is relevant with the feature *Age*. Justify.

5. **Feature Selection**

(a) (8 points) Below are descriptions of four feature selection methods. For each method, identify whether it is a *wrapper*, *filter*, or *embedded* method, and provide a brief justification for your answer.

Method A: This method evaluates features by fitting the model repeatedly with subsets of features and eliminating the least effective ones based on model performance until only a refined set of features remains.
wrapper [ ] filter [ ] embedded [ ]

Method B: This method selects features by measuring their relationship with the target variable, retaining features that surpass a specified correlation threshold independently from the model.
wrapper [ ] filter [ ] embedded [ ]

Method C: During the model training process, this method automatically retains features that contribute most directly to the model's predictions by assigning internal weights to each feature and discarding those below a weight threshold.
wrapper [ ] filter [ ] embedded [ ]

Method D: This method examines the distribution of each feature across different target classes and retains only those features that show a clear separation between classes. Features with overlapping distributions across classes are excluded.

wrapper ☐ filter ☐ embedded ☐

(b) (2 points) In a dataset with a large number of highly correlated features, the objective is to improve both model interpretability and performance. Compare the applicability of *Method A* and *Method B* for this scenario. Discuss which method may be more effective and any potential trade-offs involved.

6. **Unsupervised learning**

The *penguins dataset* includes measurements of *Adelie*, *Gentoo*, and *Chinstrap* penguin species from the Palmer Archipelago, Antarctica. The features include: *species*, *island*, *culmen (bill) length*, *culmen (bill) depth*, *flipper length*, *body mass*, and *sex*. The scientists who collected the data would like to use unsupervised learning for analysis and visualization.

The PCA plot shows the penguins projected onto two principal components (PC1 and PC2).

(a) (1 point) Briefly describe how the concept of variance is related to principal components.

(b) (2 points) Based on the plot, describe the distribution of the penguin *species* in terms of the first two principal components (PC1 and PC2). What does the clustering of the species suggest about their differences?

(c) (2 points) Comment on the separation of penguins by *sex* within each *species*. Does PCA indicate a strong variance contribution from *sex*? Justify your answer based on the plot.

(d) (2 points) If the PCA loadings for PC1 show high values for culmen length and flipper length, interpret what this implies about the differences between the species.

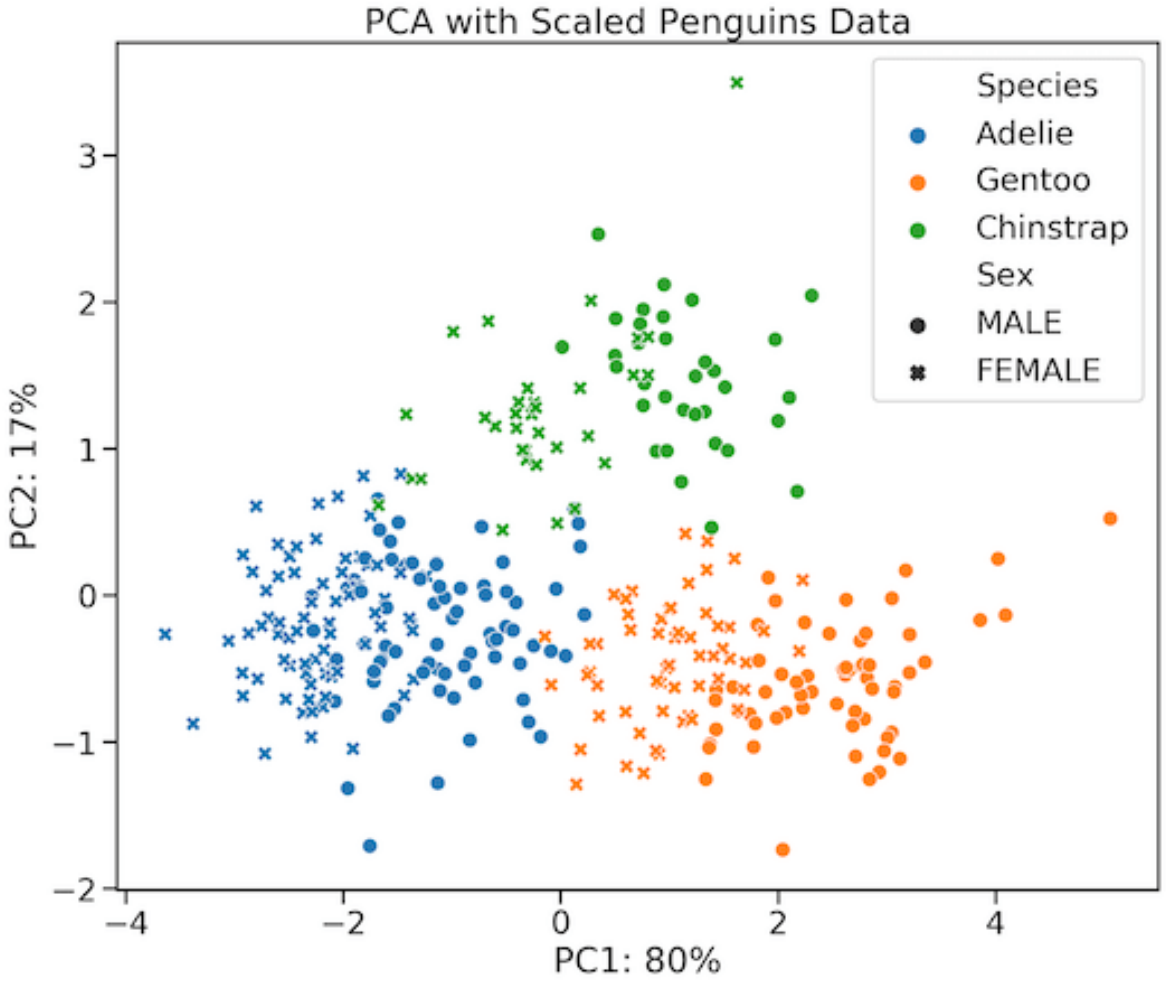


Figure 3: PCA Plot of Penguins Dataset showing clustering by species and sex.

(e) (2 points) Suppose PC2 has high loadings for culmen depth and body mass. What does this suggest about the variance captured by PC2 and how it might relate to distinguishing individuals within a species?

(f) (2 points) Given that PC1 captures 80% of the variance and PC2 captures 17%, discuss whether you think it is necessary to consider additional principal components for this dataset. What might be some reasons to consider or ignore them?

(g) (2 points) If a new feature were added to the dataset (e.g., feather length), what impact might this new feature have on the existing principal components?

7. **Hypothesis testing** This section will evaluate understanding of how to use hypothesis testing.

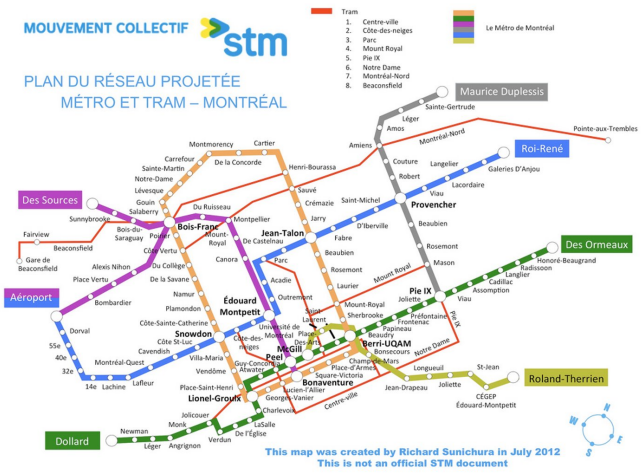


Figure 4: A drawing of the STM map.

In Figure 4 we are showing the STM map for Montreal. In this question, we want to understand if it is better to take the bus or the Metro. We have 37 samples of people taking the metro and 18 samples of people taking the bus to UdeM.

(a) (2 points) What assumptions are **sufficient** before we can use a t-test to evaluate if the data indicates taking the metro is faster than the bus?
The data is uniformly distributed ☐ The data is normally distributed ☐ The data is distributed between [0, 1) ☐ The data has equal variance ☐ There are more than 40 datapoints in each dataset ☐ There are less than 40 datapoints in each dataset ☐

(b) (1 point) What is the null hypothesis?
A variable that has not yet been assigned. ☐ Equal to 0. ☐ A result that is not sharp. ☐ Indicates two distributions look the same. ☐ There is not enough data ☐ The bus and the train take the same time ☐

(c) (2 points) What are the two quantities of the data that are compared in computing the t-value? Give two words.

(d) (2 points) How to interpret a p-value $< .05$? (Only one correct answer)

"The p-value $< .05$ indicates the probability that the null hypothesis is true is less than 5%" ☐

"A p-value of < 0.05 means that there is a 5% chance that our result is due to chance." ☐

"If the p-value $< .05$, this proves that the alternative hypothesis is true with a probability greater than 95%" ☐

"The p-value $< .05$ indicates that the probability of obtaining a result as extreme as the one observed, assuming the null hypothesis is less than 5%" ☐

(e) (1 point) If we use a t-test to check if the distribution is of the proper shape and size and we get a value of 0.7, is this value good or bad?
Good ☐, Bad ☐, Why?

(f) (1 point) Now we want to know if driving (38 samples) or taking the bike (41 samples) is a better option for getting to UdeM. If we want an effective p-value of 0.05 what p-value should we use across our t-test?
☐, Why?

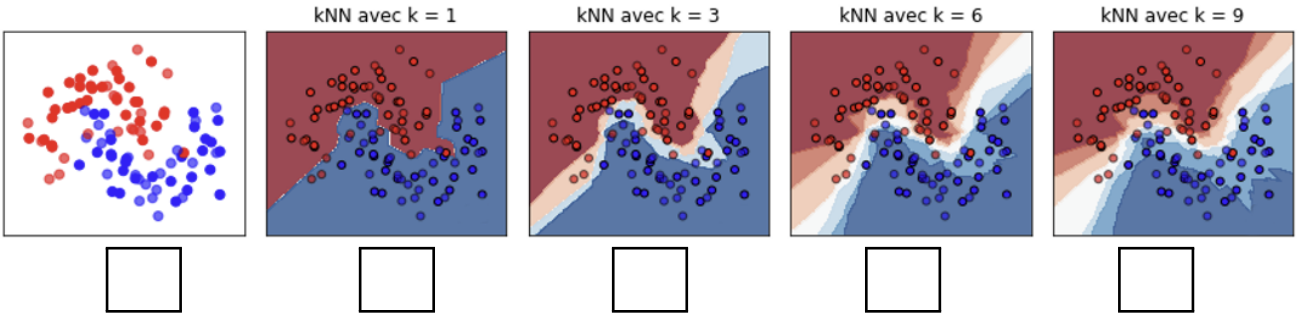
(g) (2 points) If we don't get the p-value were need on our first try of collecting data, we should try collecting data again and re-running the same test?
Yes ☐, No ☐. Why?

8. **Optimal Parameter Selection** In this section, we will work through the process of finding optimal parameters.

(a) (2 points) What is the difference between parameters and hyperparameters?

(b) (2 points) Each selection of hyperparameters influences which of the following?
The number of model parameters: ☐ final parameter values: ☐ Training time: ☐ Model performance: ☐

(c) (2 points) Select the subfigure that has the best-fit model.



☐ ☐ ☐ ☐ ☐

Figure 5: The input dataset is on the left, and the right figures show model fits for different values of k.

Why is the figure the best-fit model?

(d) (3 points) Put these steps in the correct order (write a number) to describe the optimal model section process.
Train Models: ☐ Data preparation and cleaning: ☐ Define the set hyperparameters: ☐ Define the bounds on the hyper-parameters: ☐ Deploy model: ☐

We have 3 main types of hyper-parameter optimization algorithms: (1) regular grid-based, (2) uniform random, and (3) Bayesian sampling. For each situation, indicate which algorithm is usually the best to use.

(e) (1 point) We have a configuration where we have 3 hyper-parameters, each with 40 possible ordinal settings, and we **do not** have enough compute to run all settings in parallel. ☐

(f) (1 point) We have a configuration where we have 10 hyper-parameters, each with continuous value settings, and we have a budget of 200 jobs we can run. ☐

(g) (1 point) We have a configuration where we have 3 hyper-parameters, each with 4 possible settings, and we have enough compute to run all settings in parallel. ☐

(h) (1 point) We have a configuration where we have 3 hyper-parameters, each with continuous value settings. ☐

Additional Page 1 (if necessary)

Use question and part numbers clearly if this page has to be considered

Additional Page 2 (if necessary)

Use question and part numbers clearly if this page has to be considered